

Practical No: 11

Write a Program to Build a sentiment analysis system.

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In [1]: import nltk
from nltk.sentiment import SentimentIntensityAnalyzer
import pandas as pd

nltk.download('vader_lexicon')

def analyze_sentiment(text):
    sia = SentimentIntensityAnalyzer()
    sentiment_scores = sia.polarity_scores(text)
    if sentiment_scores['compound'] >= 0.1:
        sentiment = "Positive"
    elif sentiment_scores['compound'] <= -0.1:
        sentiment = "Negative"
    else:
        sentiment = "Neutral"
    return sentiment, sentiment_scores

def analyze_sentiments(texts):
    results = []
    for text in texts:
        sentiment, scores = analyze_sentiment(text)
        results.append({
            'text': text,
            'sentiment': sentiment,
            'pos_score': scores['pos'],
            'neg_score': scores['neg'],
            'neu_score': scores['neu'],
            'compound_score': scores['compound']
        })
    return pd.DataFrame(results)

# Example usage
texts = [
    "I absolutely love this product! It's amazing!",
    "This is the worst experience I've ever had.",
    "The movie was okay, nothing special.",
    "I'm feeling pretty neutral about the whole situation.",
    "The customer service was excellent and very helpful!"
]

results_df = analyze_sentiments(texts)
print(results_df)
```

```
[nltk_data] Downloading package vader_lexicon to /root/nltk_data...
```

	text	sentiment	pos_score	\
0	I absolutely love this product! It's amazing!	Positive	0.689	
1	This is the worst experience I've ever had.	Negative	0.000	
2	The movie was okay, nothing special.	Neutral	0.233	
3	I'm feeling pretty neutral about the whole sit...	Positive	0.439	
4	The customer service was excellent and very he...	Positive	0.541	

	neg_score	neu_score	compound_score
0	0.000	0.311	0.8713
1	0.369	0.631	-0.6249
2	0.277	0.490	-0.0920
3	0.000	0.561	0.5719
4	0.000	0.459	0.7955